Softmax and Cross Entropy in Classification

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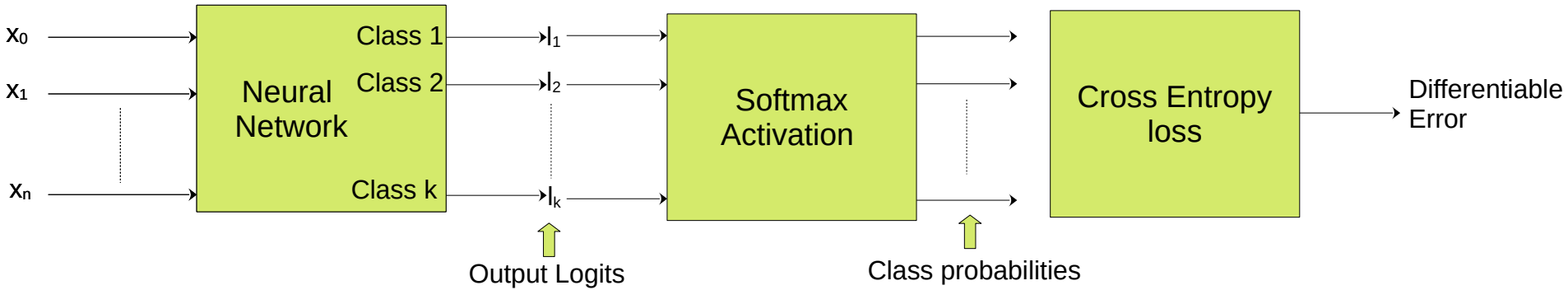
Goals:

- Explain what softmax does and why we use it
- Define cross-entropy and its gradient
- Implement the pair correctly for multi-class classification
- Avoid common pitfalls and handle class imbalance/calibration

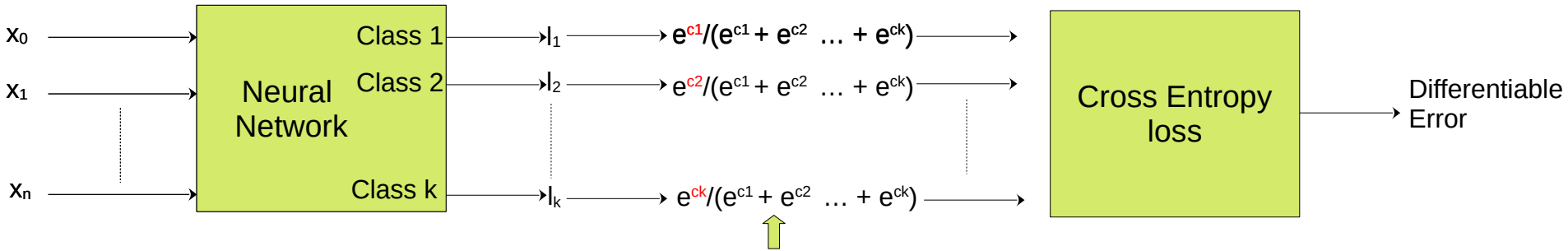
Where They Fit in a Classifier



- **Logits**: arbitrary real scores from model (unnormalized)
- Largest logit is the winner
- **Softmax** outputs class ‘probabilities’
- **Cross Entropy** how wrong we are.

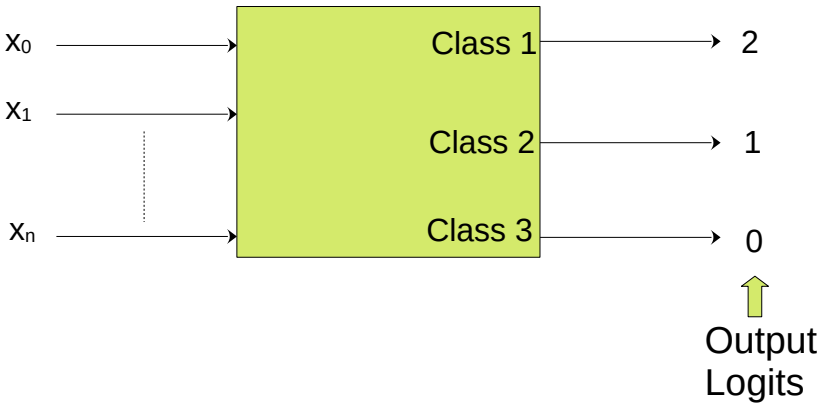
Softmax Activation

$$\frac{e^{x_i}}{\sum_{j=1}^{n} e^{x_j}}$$

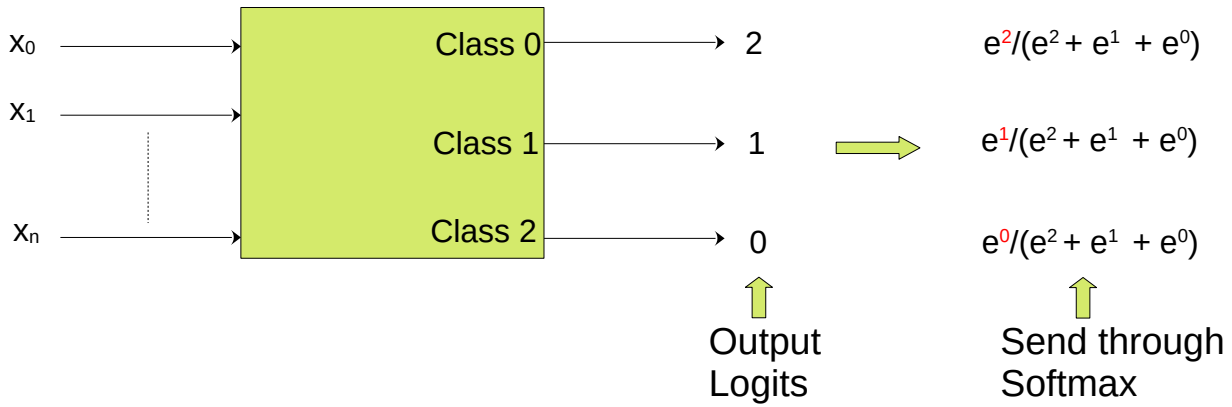


- Turns raw scores (“logits”) into a ‘probability’ like distribution (sum to 1)
- Differentiable
- Largest softmaxed logit is still largest
- In fact softmax pushes largest higher and others lower

Softmax Activation Example



Softmax Activation Example



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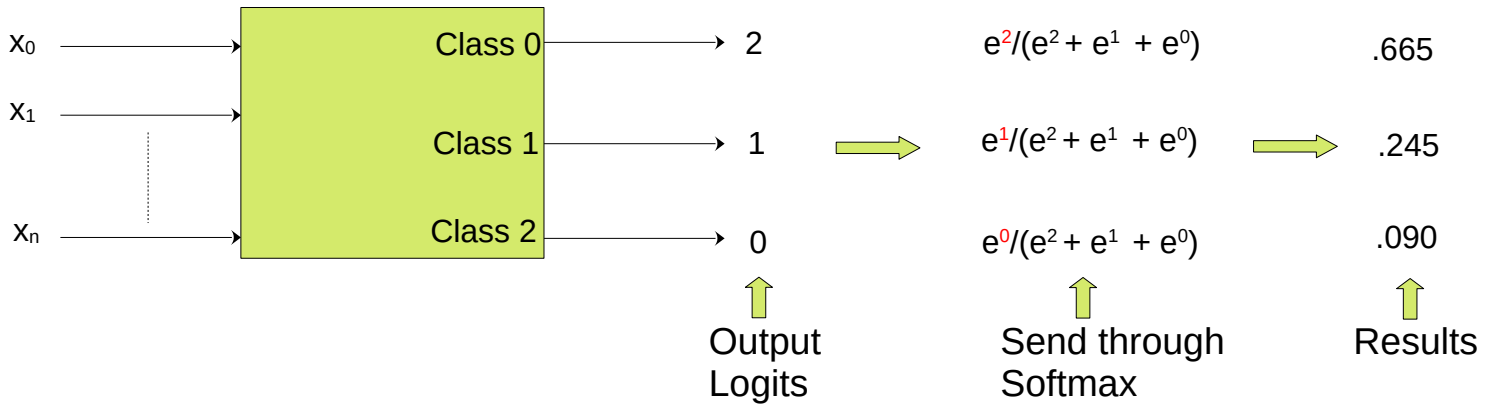
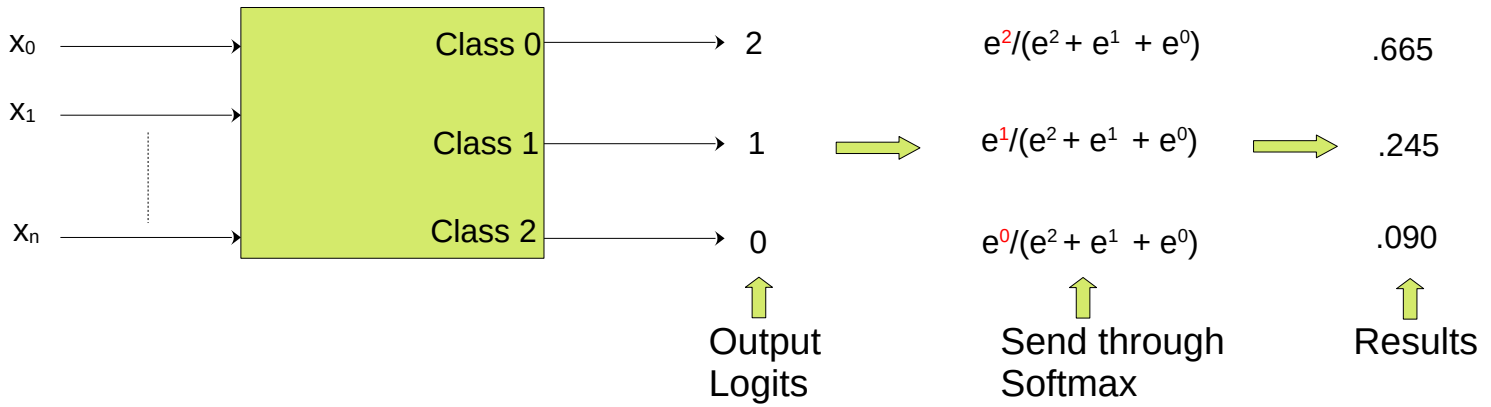
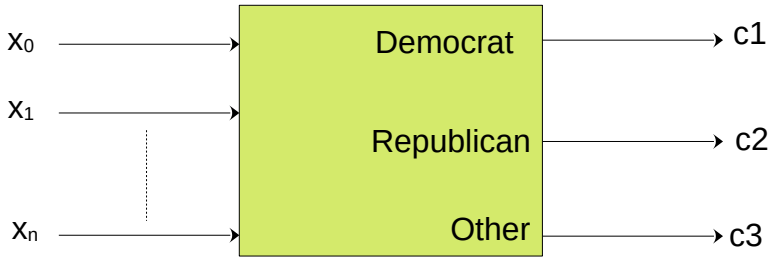


Softmax Activation Example



- .665 + .245 + .090=1
- Class 0 is still selected
- Class 0 probability is much larger than the original gap between Class0 and Class 1 suggests

Softmax Classification Example
use argmax to select predicted class



A NN with a final layer softmax activation function.
Three training rows are fed in with the following results

c1	c2	c3	targets	correct?
0.3	0.3	0.4	0 0 1 (democrat)	yes
0.3	0.4	0.3	0 1 0 (republican)	yes
0.1	0.2	0.7	1 0 0 (other)	no

It has a classification accuracy of 2/3 or 66%
It has a classification error of 1/3 or 33%
It barely gets the first 2 right and is way off on the third.

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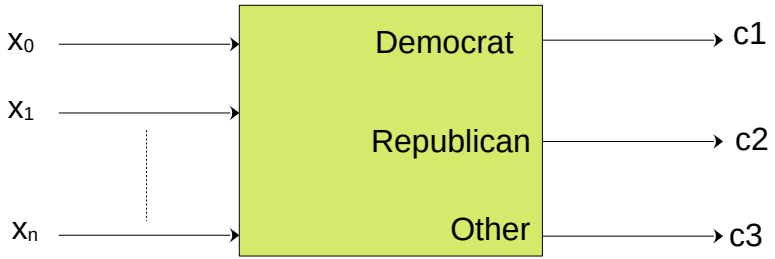
use argmax to select predicted class



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A different NN with a final layer softmax activation function.
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It nails the first 2 and barely misses the third.

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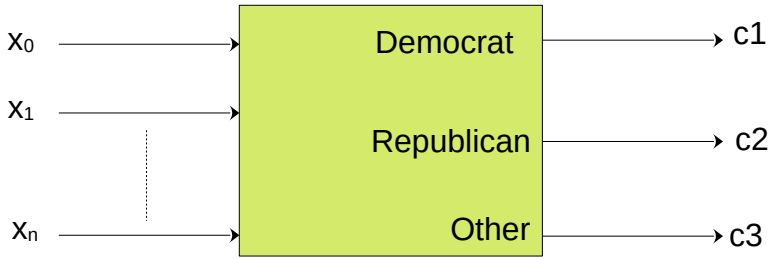

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The second network is clearly better but the metrics do not reflect that
Also, argmax is not differentiable so cannot backprop through it

Cross Entropy Classification Error

- Also called the negative log likelyhood
- Takes the negative log of the softmax score for ONLY the correct class. All other entries are ignored.
- So applying cross entropy error for the previous 2 examples;

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Row1 error=-((ln(0.3)*0) + (ln(0.3)*0) + (ln(0.4)*1)) = -ln(0.4)
Row2 error=-ln(.4)
Row3 error=-ln(.1)

Total error=-(ln(0.4) + ln(0.4) + ln(0.1)) / 3 = 1.38

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Cross Entropy Classification Error
Why it is better

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The error now reflects that the second network is much better than the first

Pytorch Cross Entropy Example

```
criterion = nn.CrossEntropyLoss()
```

```
def train(model, train_loader, test_loader, optimizer, epochs=10)
    for ep in range(1, epochs+1):
        model.train()
        run_loss, n = 0.0, 0
        for xb, yb in train_loader:
            xb, yb = xb.to(device), yb.to(device)
            optimizer.zero_grad(set_to_none=True)
            logits = model(xb)
            loss = criterion(logits, yb)
            loss.backward()
```

Note: pytorch cross entropy will automatically
apply softmax before it calculates the cross entropy

When not to use Cross Entropy

- Multi-label problems (classes are not mutually exclusive)
- Instead feed logits into BCEWithLogitsLoss

Summary

- For single label classification use cross entropy loss
- BTW it automatically applys softmax to the output logits
- For multilabel classification use BCEWithLogitsLoss